Read Me – The European Drought Impact report Inventory EDII V2.0

The data of the European Drought Impact report Inventory (EDII) archives information from text reports about negative drought impacts ("negative environmental, economic or social effects experienced as a consequence of drought"). The EDII is a research database that is a result of international collaborative work from various research projects that started with the EU FP-7 project Drought R&SPI and was followed by many others. Over time, many partners have been involved in the data collection and therefore information sources vary. Despite the best of effort for structured coding into categories and types, the styles of data entries also vary. As mentioned by various published studies, due to this history, the EDII shows a sampling bias in terms of regional differences (the project areas) and has regional and temporal gaps (no funding for collection outside project times). The investigated sources to gather reported drought impacts also have different foci. For example, the French database *Propluvia* informs people on water restrictions in drought periods that are classified in the EDII as impacts on waterborne transportation, irrigation etc. and the *DMCSEE bulletins* focus on agriculture. In contrast, other sources such as the Swiss *DROUGHT-CH* are broader 'media report collections' and were therefore classified into many different impact categories. Subsequently, the resulting report-based impact data collection of the EDII is influenced by national priorities. Different collection efforts, such as regional surveys in Germany based on interviews and questionnaires lead to particularly strong regional sampling biases (e.g. hydropower or public water supply). As the entries to the database are based on reports, a particular impact may be reported several times by different reports. Consequently, any analysis undertaken with this research data needs (1) a defined and filtered data sample and (2) a proper quality check and/or correction.

The original guidelines on how to derive drought impact data from text reports for the EDII are attached as Guidelines_EDII.pdf.

The EDII data is attached as EDII_DataV2.0.txt and EDII_DataV2.0.xlsx.

For more details on the database creation and content read:

- Stahl, K., Kohn, I., Blauhut, V., Urquijo, J., De Stefano, L., Acácio, V., Dias, S., Stagge, J. H., Tallaksen, L. M., Kampragou, E., van Loon, A. F., Barker, L. J., Melsen, L. A., Bifulco, C., Musolino, D., de Carli, A., Massarutto, A., Assimacopoulos, D., and van Lanen, H. A. J.: Impacts of European drought events: insights from an international database of text-based reports, Nat. Hazards Earth Syst. Sci., 16, 801–819, https://doi.org/10.5194/nhess-16-801-2016, 2016.

The Alpine Drought Impact report Inventory $EDII_{ALPS}$ (DOI: 10.6094/UNIFR/230219) is a sub database of the EDII focusing on the "Alpine Space" introduced by the EU-funded program of the same name. The $EDII_{ALPS}$ has been developed during the project Alpine Drought Observatory (https://ado.eurac.edu/).

For more details read:

- Stephan, R., Erfurt, M., Terzi, S., Žun, M., Kristan, B., Haslinger, K., and Stahl, K.: An inventory of Alpine drought impact reports to explore past droughts in a mountain region, Nat. Hazards Earth Syst. Sci, 21.

Some examples of studies that carefully applied information of the EDII database reports for specific statistical analyses are:

- Bachmair, S., Svensson, C., Prosdocimi, I., Hannaford, J., Stahl, K., 2017. Developing drought impact functions for drought risk management. NAT HAZARD EARTH SYS 17, 1947–1960, https://doi.org/10.5194/nhess-17-1947-2017.
- Bachmair, S., Svensson, C., Hannaford, J., Barker, L.J. and Stahl, K., 2016. A quantitative analysis to objectively appraise drought indicators and model drought impacts. Hydrology and Earth System Sciences, 20(7), pp.2589-2609.
- Blauhut, V., Gudmundsson, L., Stahl, K., 2015. Towards pan-European drought risk maps: Quantifying the link between drought indices and reported drought impacts. ENVIRON RES LETT 10, 14008, https://doi.org/10.1088/1748-9326/10/1/014008
- Blauhut, V., Stahl, K., Stagge, J.H., Tallaksen, L.M., De Stefano, L. and Vogt, J., 2016. Estimating drought risk across Europe from reported drought impacts, drought indices, and vulnerability factors. Hydrology and Earth System Sciences, 20(7), pp.2779-2800.
- Erfurt, M., Skiadaresis, G., Tijdeman, E., Blauhut, V., Bauhus, J., Glaser, R., Schwarz, J., Tegel, W. and Stahl, K., 2020. A multidisciplinary drought catalogue for southwestern Germany dating back to 1801. Natural Hazards and Earth System Sciences, 20(11), pp.2979-2995.
- Stagge, J.H., Kohn, I., Tallaksen, L.M., Stahl, K., 2015. Modeling drought impact occurrence based on meteorological drought indices in Europe. J HYDROL 530, 37–50, https://doi.org/10.1016/j.jhydrol.2015.09.039
- Sutanto, S.J., van der Weert, M., Blauhut, V. and Van Lanen, H.A., 2020. Skill of large-scale seasonal drought impact forecasts. Natural Hazards and Earth System Sciences, 20(6), pp.1595-1608.
- Tijdeman, E., Blauhut, V., Stoelzle, M., Menzel, L. and Stahl, K., 2022. Different drought types and the spatial variability in their hazard, impact, and propagation characteristics. Natural Hazards and Earth System Sciences, 22(6), pp.2099-2116.

FILE FORMAT (Missing value code = NA):

Bold labels indicate mandatory information to archive and classify reported drought impacts in the EDII. The presented dataset was checked for missing entries, however to ensure completeness further checks should be applied before using for statistical analyses. Information of non-bold labels are provided only if the source report mentioned such additional information.

The presented data are unique regarding the ID. A single drought impact (reference) report can report on different countries (Country), regions (NUTS1_ID, NUTS2_ID, NUTS3_ID), impact categories (Impact_category) and the impact subtypes (Impact_subtype). Each ID corresponds to a single region and a single impact subtype, meaning an impact description (Impact_description) can be repeated, if several regions, impact categories or impact subtypes are assigned. The EDII uses the NUTS 2016 classification (https://ec.europa.eu/eurostat/web/nuts/background) for the spatial reference of the reported drought impacts.

- **ID**: Entry id to enable tracing back the entry in the EDII.
- **Year_start**: Year of start of impact occurrence.
- Season_start: Season of start of impact occurrence.
- Month_start: Month of start of impact occurrence.
- Year_end: Year of end of impact occurrence.
- Season_end: Season of end of impact occurrence.
- Month_end: Month of end of impact occurrence.
- Related_drought_event: Related "well-known" drought event leading to the reported impact.
- **Country**: CNTR_Code of the country, for which the impact was reported.
- NUTS1_ID: NUTS 1 classification of the region, for which the impact was reported.
- NUTS2_ID: NUTS 2 classification of the region, for which the impact was reported.
- NUTS3_ID: NUTS 3 classification of the region, for which the impact was reported.
- Location: Location, for which the impact was reported
- Rivers_streams: Rivers or streams, for which the impact was reported.
- Lakes_reservoirs: Lakes or reservoirs, for which the impact was reported.
- Latitude: Latitude coordinates of the reported impact.
- Longitude: Longitude coordinates of the reported impact.
- Alititude: Altitudes, for which the impact was reported.
- **Impact_category**: The category, to which the reported impact was classified. Impact categories are listed below.
- **Impact_subtype**: The subtype, to which the reported impact was classified. Impact subtypes are listed below.
- **Impact_description**: Short text summary of the reported impact according the original source (free text written in English by the respective contributor).
- **Information_source:** Source of information of the underlying entry. Source types are listed below.
- **Reference**: Reference that is as complete as possible (ideally including the name of the author, year, title of report, weblink)
- People_affected: Affected people by the reported impacts as provided by the source report.
- Area_affected: Affected area by the reported impact as provided by the source report.
- Economic_loss: Economic loss from the reported impact as provided by the source report.

- Associated_impact: indirect (secondary, tertiary…) drought impacts associated with the reported impact. Suitable types are the same as for direct impacts (listed below).
- Associated_impact_description: Summary of the associated impact.
- Response_measure: Type of response measure related to the reported impact. Suitable types are listed below and explained in deStefano et al., 2012.
- Level_of_measure: Administrative level of response measure from local activities to international aid. Suitable levels are listed below and explained in deStefano et al., 2012.
- Response_description: Summary of the applied drought response measure (written by the contributor based on the information in the source report).
- Related_costs: Any information on the costs of the response measures if available (as specified in the report source).

Impact categories and impact subtypes

ID and Impact category	**Impact subtype** *ID*	*Description*
1. Agriculture and livestock farming	1.1	Reduced productivity of annual crop cultivation: crop losses, damage to crop quality or crop failure due to dieback, premature ripening, drought-induced pest infestations or diseases etc.
	1.2	Reduced productivity of permanent crop cultivation
	1.3	Agricultural yield losses ≥ 30% of normal production (EU compensation treshold)
	1.4	Reduced availability of irrigation water
	1.5	Reduced productivity of livestock farming (e.g. reduced yields or quality of milk, reduced stock weights)
	1.6	Forced reduction of stock(early selling/slaughtering)
	1.7	Regional shortage of feed/water for livestock
	1.8	Other
	1.9	Increased costs/economic losses
2. Forestry	2.1	Reduced tree growth and vitality
	2.2	Decrease in annual non-timber products from forest trees (e.g. cork, pine nuts, mushrooms, berries, etc.) (please specify which kind of product)
	2.3	Increased occurrence of water stress indicators and damage symptoms (e.g. premature ripening, seasoning checks, defoliation, worsened crown conditions etc.) (please specify forest type/tree species in the description field!)
	2.4	Increase of pest/disease attacks on trees (please specify species in the description field!)
	2.5	Increased dieback of trees (please specify tree species in the description field!)
	2.6	Increased dieback of planted tree seedlings (in nurseries or afforested area)
	2.7	Damage to short rotation forestry plantations (energy forestry)
	2.8	Other
	2.9	Increased costs/economic losses
3. Freshwater aquaculture and fisheries	3.1	Reduced (freshwater) fishery production (please specify fish species in the description field)
	3.2	Reduced aquaculture production (please specify fish species in the description field)
	3.3	Other
	3.4	Increased costs/economic losses
4. Energy and industry	4.1	Reduced hydropower production
	4.2	Impaired production/shut down of thermal/nuclear power plants (due to a lack of cooling water and/or environmental legislation for discharges into streams)
	4.3	Restriction/disruption of industrial production process (due to a lack of process water and/or environmental legislation/restrictions for discharges into streams)

	4.4	Other
	4.5	Increased costs/economic losses
5. Waterborne transportation	5.1	Impaired navigability of streams (reduction of load, increased need of interim storage of goods at ports)
	5.2	Stream closed for navigation
	5.3	Other
	5.4	Increased costs/economic losses
6. Tourism and recreation	6.1	Reduced number of short-stay-tourists
	6.2	Reduced number of long-stay-tourists
	6.3	Sport/recreation facilities affected by a lack of water
	6.4	Impaired use/navigability of surface waters for water sport activities (including bans)
	6.5	Other
	6.6	Increased costs/economic losses
	6.7	Impaired use of ski slopes and cross-country ski trails due to lack of snow/water (e.g. limited snow production, impaired grooming)
7. Public water supply	7.1	Local water supply shortage / problems (drying up of springs/wells, reservoirs, streams)
	7.2	Regional/region-wide water supply shortage/problems (drying up of springs/wells, reservoirs, streams)
	7.3	Bans on domestic and public water use (e.g. car washing, watering the lawn/garden, irrigation of sport fields, filling of swimming pools)
	7.4	Limitations in water supply to households in rural areas (supply cuts, need to ensure water supply by emergency actions)
	7.5	Limitations in water supply to households in urban areas (supply cuts, need to ensure water supply by emergency actions)
	7.6	Other
	7.7	Increased costs/economic losses
8. Water quality	8.1	Increased temperature in surface waters (close to or exceeding critical values)
	8.2	(Temporary) water quality deterioration/problems of surface waters(natural & manmade); e.g. significant change of physio-chemical indicators, increased concentrations of pollutants, decreased oxygen saturation levels, eutrophication, algal bloom)
	8.3	(Temporary) impairment of ecological status of surface waters (according to EU Water Framework Directive)
	8.4	(Temporary) impairment of chemical status of surface waters (according to EU Water Framework Directive)
	8.5	Increased salinity of surface waters (saltwater intrusion and estuarine effects)
	8.6	Problems with groundwater quality
	8.7	Increased salinity of groundwater
	8.8	Problems with drinking water quality (e.g., increased treatment, violation of standards)
	8.9	Problems with bathing water quality
	8.10	Problems with irrigation water quality
	8.11	Problems with water quality for use in industrial production processes
	8.12	Other
	8.13	Increased costs/economic losses
9. Freshwater ecosystems: habitats, plants and wildlife	9.1	Increased mortality of aquatic species (specify species (latin term) and state whether a rare/endangered/protected species is concerned in the description field)
	9.2	Increased species concentration near water
	9.3	Migration and concentration (loss of wildlife in some areas and too many in others)
	9.4	Increased populations of invasive (exotic) aquatic species

	9.5	Observation of adverse impacts on populations of rare/endangered (protected) riparian species
	9.6	Observation of adverse impacts on populations of rare/endangered (protected) species of wetlands
	9.7	Loss of biodiversity (decrease in species diversity)
	9.8	Danger for or actual violation of minimum flow or environmental flow requirements
	9.9	Drying up of shallow water areas, weed growth or algae bloom
	9.10	Drying up of perennial stream sections
	9.11	Drying up of lakes and reservoirs (which have a habitat function)
	9.12	(Mid-/Long-term) deterioration of wetlands
	9.13	Irreversible deterioration/loss of wetlands
	9.14	Other
	9.15	Increased costs/economic losses
10. Terrestrial ecosystems: habitats, plants and wildlife	10.1	Increased species mortality (specify species (latin term) and state whether a rare/endangered/protected species is concerned)
	10.2	Changes in species biology/ecology
	10.3	Loss of biodiversity (decrease in species diversity)
	10.4	Shift in species composition
	10.5	Reduced plant growth
	10.6	(Mid-/Long-term) deterioration of habitats
	10.7	Irreversible deterioration/loss of habitats
	10.8	Lack of feed/water for terrestrial wildlife
	10.9	Increased attacks of pests and diseases
	10.10	Increased contact of wild animals under stress (shortage/lack of feed and water) with humans/human settlements
	10.11	Other
	10.12	Increased costs/economic losses
11. Soil system	11.1	Drought-related erosion processes (loss of soil fertility)
	11.2	Structural damage to private property due to soil subsidence/shrinkage
	11.3	Structural damages on infrastructures due to soil subsidence/shrinkage
	11.4	Other
	11.5	Increased costs/economic losses
12. Wildfires	12.1	Increased burned area
	12.2	Increased number of wildfires
	12.3	Increased severity of wildfires
	12.5	Increased costs/economic losses
	12.6	Other
13. Air quality	13.1	Air quality pollution effects/problems (dust bowl effect, wildfires, substitution of hydropower production by fossil energy)
	13.2	Other
	13.3	Increases costs/economic losses
14. Human health and public safety	14.1	Heat stress problems (if drought is associated with a heatwave)
	14.2	Increased respiratory ailments (heat wave and air quality)
	14.3	Excess mortality during heat waves
	14.4	Drought induced public-safety issues (e.g. increased risk of structural damages)
	14.5	Other
	14.6	Increases costs/economic losses
15. Conflicts	15.1	Water allocation conflicts – international
	15.2	Regional/local user conflicts
	15.3	Other
	14.4	Increases costs/economic losses

Information sources

ID	Information source
1	Database
2	Scientific Article
3	Book (Book Section)
4	Newspaper Article
5	Governmental Report (national and regional agencies/organization)
6	(Other) Government Document
7	River Basin Organization Report
8	Report (by NGO)
9	Press Release by stakeholders or agencies
10	Report (private sector like insurances)
11	Thesis
12	Pamphlet
13	Personal Observation
14	Web Page
15	Map
16	Other
17	Interview / Questionnaire
18	Tambora

Response measures (selection list)

- Preventive and strategic measures (developed and used under normal (pre-alert) conditions, aim to reinforce the structural system to increase its response capacity towards droughts)
- Education and awareness campaigns
- Water demand management (serve to control or influence the amount of water used - to decrease water use before, during or after drought periods - including, for instance, improving water use efficiency, water saving measures in different sectors, prioritization, reductions and restrictions of uses)
- Water supply management (aim at increasing the availability of water to be used)
- Environmental measures (aim at mitigating the impacts of drought on ecosystems)
- Recovery measures (compensation schemes, ecosystem restoration) (deactivation of adopted measures and the activation of restoration measures for the systems affected by drought)
- Others

Administrative levels of response measure (selection list)

- local activities
- regional / national measures
- international aid
- no information

For more information on response measures read:

- De Stefano et al., 2012. Policy and Drought Responses – Case Study Scale. DROUGHT-R&SPI Technical Report 4. http://www.eu-drought.org/technicalreports/10821986/DROUGHT-R-SPI-Technical-Report-No-4-Policy-and-Drought- Responses-Case-Study-scale